Review Sheet 4: Some More Bets[1]

1. Banking Crisis

You are in the unfortunate but well-paid position of being Chief Investment Officer at a leading hedge fund. While your fund has been performing well, the contagion and fear in capital markets is getting to your investors. This is leading to huge redemptions from your fund. You must plan to meet these redemptions at the lowest costs. You and your economic team employ the Quaker method of estimating these redemptions and predict that over the course of the next month, the expected redemptions will be US$ 300 Million. The standard deviation of this estimate is US$100 million. Unfortunately, all your hedge fund assets are illiquid and cannot be disposed for the next year. You must thus go to the capital markets for raising additional funds.
You have two sources of financing that you are planning to use simultaneously:
Source 1: You can borrow from the government sponsored TARP (The-Augmentation-of-Rich-People) program. This facility charges a usurious 25% annual rate of interest. Further, it can be availed only if a request is made far in advance.
Source 2: Alternately, when TARP funding comes short, you can borrow from the overnight credit markets. You can use this facility anytime and it provides funding immediately. Unfortunately, the rate of interest charged is 40% per annum.

a) ***How much money would you borrow from the TARP funds?****

b) ***What are your annualized capital costs if you employ the solution prescribed in part A?***

c) ***One of your classmates at INSEAD is in exactly the same financial position as you. While he runs an identical fund, most of his clients are located in a different geography. He calls you up and proposes merging your two funds. He somewhat remembers the POM class***

[1] Problems are rated for difficulty: * = Relatively easy, for first review of concepts; ** = Challenging, assumes good understanding of the concepts; *** = Advanced, assumes full understanding of the concepts and creativity in application.

and suggests that merging would reduce the above mentioned capital costs for both funds. Do you agree? Justify your answer in no more than 100 words. *

d) ***The crisis is getting worse. The overnight credit markets have now completely frozen up. You no longer care about minimizing your capital costs. All you care about is the odds of your survival, or the probability that your fund will not be able to meet redemptions. What is the minimum amount of money you need to borrow from TARP funds to ensure that you will be able to meet all redemption requests with a 95% probability?****

2. Energy Economics

A massive natural gas field has been discovered in the beautiful Central Asian Republic of Gasistan. The gas field provides a limitless supply of gas that has the potential to transform the lives of the people (and rulers) of Gasistan. However, to realize this prosperity the National Gas Corporation of Gasistan (NGCG) must develop facilities to transport the plentiful natural gas to energy-hungry markets. The geological makeup of this field dictates that gas that is produced must be transported to markets or it is lost. Assume that for the time-frame of the analysis, the field can produce any quantity of gas that can be transported.

You have been hired as a consultant to the NGCG. Based on your research, you have outlined two potential ways to transport gas to somewhat differing markets.

1. **Pipeline:** *A pipeline can be built to transport a fixed quantity of gas to European markets. Daily demand in the European markets that can be accessed by the proposed pipeline is distributed Normally with a mean of 600 Million Cubic Feet (mcf) and a standard deviation of 600 mcf. The low operation costs of the pipeline ensure that every mcf of gas that can be sold in European markets, NGCG earns a profit of $10,000 (taking into account* ***operating costs****, i.e., production, transportation and all other operating costs). Building a pipeline involves a heavy capital investment. Based on the availability of capital and the sovereign guarantees available to the NGCG, you estimate that the daily* **capital cost** *of one mcf of transportation capacity will be $5,000. In other words, if a pipeline with daily capacity of 1,000 mcf is built, the daily capital cost would be $5 million.*
2. **Liquefying and shipping terminal:** *Natural gas may also be liquefied and can then be sent via special sea-going tankers to a variety of global markets. Daily Demand in the markets that can economically be accessed by sea-going vessels is distributed normally with the same mean as daily demand in European markets, 600mcf. However, the diversity of markets that can be accessed by sea-going vessels limits the impacts of region-specific demand shocks and the standard deviation of the accessible markets is only 200 mcf. Further, you estimate that every mcf of gas that can be sold in these markets will earn a profit of $9,000. Finally, you estimate that the daily capital cost of one mcf of liquefying and transportation capacity will be $6,000.*

a) If you decide to build a pipeline, what is the capacity of the pipeline that maximizes the expected profits?*

b) If you decide to build a pipeline with the capacity calculated in A., what would be the average utilization of that pipeline? *

c) With the pipeline option, what are the daily profits? (taking into account the capital costs)*

d) ***If you decide to build the liquefying facility, what is the profit maximizing capacity of the liquefying and shipping terminal? ****

e) ***If you decide to build the liquefying facility with the capacity calculated in D, what would be the utilization of that facility?****

f) ***With the liquefying option, what are the daily profits? (taking into account the capital costs)****

g) ***Based on your computation of daily profits, which of the two options would you chose. Qualitatively, highlight the key difference between the two options that drives your recommendation. ****

3. Land's End**

Geoff Gullo owns a small firm that manufactures "Gullo Sunglasses". He has the opportunity to sell a particular seasonal model to Land's End. Geoff offers Land's End two purchasing options:
Option 1: Geoff offers to set his price at $65 and agrees to credit Land's End $53 for each unit Land's End returns to Geoff at the end of the season (because these units did not sell). Since styles change each year, there is essentially no value in the returned merchandise.
Option 2: Geoff offers a price of $55 for each unit, but returns are no longer accepted. In this case, Land's End throws out unsold units at the end of the season.
The season's demand for this model is Normal with mean 200 and standard deviation 125. Land's end will sell those sunglasses for $100 each. Geoff's production cost is $25.

a) How much would Land's End buy if they chose Option 1?*

b) How much would Land's End buy is they chose Option 2?*

c) Which option will Land's End choose?*

d) ***Suppose Land's End chooses option 1 and orders 275 units. What is Geoff Gullo's expected profit?****